

Log File Analyzer for Intrusion Detection

Introduction

Log monitoring is an essential part of cybersecurity operations. Security analysts monitor logs to detect suspicious activities, investigate incidents, and identify potential threats. This project focuses on analyzing Linux authentication logs to detect brute-force attacks and unauthorized access attempts.

Abstract

The Log File Analyzer for Intrusion Detection is a Python-based security monitoring tool designed to analyze Linux authentication logs. The tool detects failed login attempts, extracts suspicious IP addresses, identifies brute-force attacks, assigns risk levels, and generates security reports. It also visualizes attack patterns through graphs and exports findings for further analysis.

Tools Used

- Python
- Pandas
- Matplotlib
- Regex
- Linux Authentication Logs

Log File Analyzer for Intrusion Detection

Steps Involved

1. Collected authentication logs.
2. Parsed log entries using Python.
3. Detected failed login attempts.
4. Extracted attacker IP addresses.
5. Classified attacks based on severity.
6. Generated incident reports.
7. Exported findings to CSV.
8. Created attack visualization graphs.

Conclusion

This project demonstrates how log analysis can be used to identify suspicious activities and potential security incidents. The tool provides a simple intrusion detection mechanism that helps security analysts detect brute-force attacks and unauthorized access attempts. The project enhanced understanding of SOC operations, log monitoring, and incident detection techniques.